

Motion-consistent 4D reconstruction and segmentation for sparse short-axis CMR: a public methods extension of ReconSeg3D

Working manuscript (methods)

Status: Publication-oriented draft. Quantitative panels report **demo-regime pipeline metrics only**. Subject-level public-benchmark tables remain **to be completed** after licensed dataset mounts.

Target genre: Nature-family / Medical Image Analysis-style methods · English

Code availability: <https://github.com/Coucou2016/ReconSeg3D-cardiac-rec>

Abstract

Sparse short-axis (SA) cine cardiac magnetic resonance (CMR) undersamples three-dimensional anatomy across the cardiac cycle, motivating dense spatiotemporal reconstruction before structure-aware analysis. Gao et al. (*npj Digital Medicine*, 2026) showed that reconstructed bi-ventricular cine volumes can support long-horizon major adverse cardiovascular event (MACE) prediction on a large private acute myocardial infarction (AMI) cohort; that clinical discrimination figure is tied to private data and training and is **not claimed here**. We present a public, motion-consistent four-dimensional (4D) methods extension of a lightweight ReconSeg3D-style stack: per-frame volumetric decoding, differentiable warp regularization, and an **image-cycle** consistency loss that penalizes intensity residual after forward and backward warps—distinct from coordinate-composed inverse-consistent flow. The stack is trained jointly with multi-structure segmentation and optional phenotype or risk heads. HeartTTable-style multimodal fusion is retained only as **HeartTTable-lite** (a single class token attending to concatenated spatial, temporal, and tabular keys) and reported as a fusion ablation. On controlled demo-scale public-proxy tensors we verify end-to-end logging of reconstruction (PSNR/MAE), segmentation Dice, and motion proxies (warp L1, image-cycle error, ejection-fraction proxy). Local challenge-named folders contain synthetic placeholders rather than licensed ACDC, MM-WHS, or EMIDEC volumes; real subject-level tables remain to be completed. We do not claim private-AMI MACE parity.

Keywords: cardiac MRI; 4D reconstruction; motion consistency; multi-task learning; public benchmarks; methods reproducibility

1. Introduction

Cardiac cine magnetic resonance imaging (CMR) is typically acquired as a stack of two-dimensional short-axis (SA) slices. Slice gaps, through-plane motion, and temporal undersampling leave a sparse observation of bi-ventricular anatomy. Dense three-dimensional (3D) reconstruction across the cardiac cycle—effectively a four-dimensional (4D) volume—can improve anatomical continuity for segmentation and downstream phenotyping.

Gao et al. (*npj Digital Medicine*, 2026) introduced ReconSeg3D to reconstruct temporally resolved bi-ventricular volumes from SA stacks and HeartTTable to fuse imaging with tabular clinical variables for five-year MACE after percutaneous

coronary intervention in AMI^[1]. That study’s headline discrimination metrics are tied to a private cohort of thousands of patients and are therefore out of scope for any public reimplementations that lack those data.

Gap. Lightweight public reimplementations often broadcast a single reconstructed volume across time or omit explicit motion constraints, which weakens claims about spatiotemporal fidelity even when reconstruction and segmentation heads are present. Concurrent lines of work emphasize whole-sequence continuity for 4D segmentation^[5], joint recon–motion–seg unrolling^[3], and mesh- or tetrahedra-based 4D heart recovery from sparse views^{[7][8]}, but a compact, auditable ReconSeg3D-style training contract with named image-cycle regularization and an honest multimodal ablation boundary remains useful for reproducible methods research.

Contribution. We provide and evaluate a motion-consistent 4D training stack on public proxies:

1. Per-frame 3D reconstruction with temporal mixing (default path; broadcast baseline retained for ablation).
2. Motion-network warp losses plus image-cycle consistency, with optional left-/right-ventricular volume-curve smoothing and an ejection-fraction (EF) proxy.
3. Compact segmentation and phenotype/risk heads suitable for ACDC-style phenotype proxies when real data are available.
4. HeartTTable-lite as a fusion ablation only—not a claim of pairwise three-modality class-token cross-attention parity.
5. An explicit claim boundary separating demo-regime pipeline verification from subject-level public benchmarks and from private AMI survival evaluation.

2. Related work

Reconstruction. Deep learning has become central to accelerated and robust MRI reconstruction^[2]. Cardiac applications increasingly couple reconstruction with motion or segmentation rather than treating them in isolation.

Joint recon–motion–seg. Unified frameworks that estimate groupwise motion while refining reconstruction and segmentation demonstrate benefits of inter-task dependency^[3]. Decoupled motion–shape models address sparse-slice 4D myocardium reconstruction^[4]. Memory-based networks improve whole-sequence 4D cine segmentation by exploiting spatiotemporal continuity^[5]. Complementary mesh and deformable-tetrahedra approaches recover 4D cardiac geometry from multi-view or sparse intraoperative observations^{[7][8]}.

Multimodal survival. Neural Cox models such as DeepSurv generalize linear proportional-hazards risk scores^[6]. HeartTTable combines spatiotemporal imaging tokens with tabular variables via transformer cross-attention on private AMI data^[1]. Our public codebase exposes a Cox loss API and a lite fusion module but does not reproduce that clinical study.

Positioning. Relative to Gao et al.^[1], this draft is a methods extension focused on motion-consistent training and public-proxy evaluation, not a clinical outcome paper. Relative to Qian et al.^[3], Ye et al.^[5], and recent 4D mesh/tetrahedra recoveries^{[7][8]}, we do not claim superior Dice or mesh accuracy on challenge leaderboards; we claim a transparent lightweight contract—per-frame reconstruction, warp and image-cycle regularization, and a lite fusion ablation—with reproducible demo-regime logs.

3. Methods

3.1 Problem setup

Let $(x \in \mathbb{R}^{B \times C \times T \times D \times H \times W})$ denote a cine volume batch (channels (C) , frames (T) , depth (D) , height (H) , width (W)). Sparse SA sampling is simulated by zeroing unobserved slices along the through-plane axis. The model predicts a dense reconstruction $(\hat{x})$, optional displacement fields for warping consecutive frames, multi-class segmentations (left ventricle, right ventricle, myocardium; optional scar), and task heads for phenotype classification, binary event prediction on synthetic labels, or Cox partial likelihood when event times are available.

3.2 Per-frame reconstruction

Unlike broadcasting a single decoded volume across (T) , the default path encodes each frame with a compact 3D convolutional backbone (optional transformer bottleneck) and decodes per-frame volumes. Temporal mixing couples neighboring frames without collapsing the time axis. Compact demonstration grids (for example $(D \times H \times W = 16 \times 32 \times 32)$ with $(T=8)$) are used for pipeline verification; paper-scale grids remain configuration options for future subject-level experiments.

3.3 Motion consistency

A motion network predicts voxel displacements $((d_z, d_y, d_x))$. Displacements are scaled to normalized grid coordinates with dimension guards for singleton axes. We regularize:

- **Warp intensity error** between warped source and target intensities (L1).
- **Image-cycle error**: compose forward and backward warps on **image intensities** and penalize deviation from the original frame. This is distinct from inverse-consistent composition of flow fields in classical registration; we name the term image-cycle to avoid reviewer misreading.
- Optional **volume-curve** smoothness on fractional left-/right-ventricular volumes and an **EF proxy** derived from end-diastolic and end-systolic fractional volumes.

3.4 Segmentation and task heads

A compact 3D U-Net–style head predicts multi-class masks. Phenotype (ACDC-style), binary event, or Cox risk heads can be enabled by configuration. Discrimination metrics are pooled at the epoch level; tiny synthetic splits can yield unstable area-under-curve or concordance estimates and must not be over-interpreted as clinical performance.

3.5 HeartTTable-lite (ablation)

When multimodal fusion is enabled in the lite setting, a single class token attends over concatenated spatial, temporal, and tabular key–value streams. This design is intentionally weaker than the original three class-token pairwise cross-attention module and is reported only as a fusion ablation against a concatenation–multilayer-perceptron baseline.

3.6 Training and evaluation protocol

Demonstration configurations use compact spatial grids and short training schedules for end-to-end metric logging. Public loaders may synthesize placeholder volumes when licensed roots are absent. Subject-level cross-validation, physical-space surface-distance metrics, and calibration plots are specified for future tables after licensed mounts and are marked incomplete in the present draft.

4. Experiments

4.1 Datasets and honesty inventory

Dataset	Role	Status in this draft
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Synthetic / auto-fake tensors	Pipeline verification	**Used** for all quantitative panels below
ACDC (challenge)	Reconstruction, segmentation, phenotype proxy	**Not mounted.** Local challenge-named folders hold demo-scale synthetic NIfTI placeholders (eight subjects; compact $(32 \times 32 \times 16)$ volumes). Official download requires institutional registration
MM-WHS	Static segmentation / table fusion	Demo-scale placeholders; motion off / $(T=1)$ recommended when real data arrive
EMIDEC	Scar / infarct proxy	Demo-scale placeholders; subject-level tables incomplete
Private AMI	Five-year MACE	Out of scope (example manifest only; no cohort redistribution)

4.2 Ablations

We compare: broadcast reconstruction baseline; per-frame decoding without motion loss weights; per-frame decoding with warp and image-cycle terms; concatenation versus HeartTTable-lite fusion; and a phenotype-task configuration. Summaries are aggregated from measured demo-regime ablation logs.

4.3 Metrics

Reconstruction: peak signal-to-noise ratio (PSNR), global structural-similarity proxy, mean absolute error (MAE). Segmentation: per-class Dice; optional subsampled Hausdorff-95 (approximate on compact grids). Motion: warp L1, image-cycle error, EF proxy. Phenotype / event / concordance indices when applicable—demo values are not clinical performance.

5. Results

5.1 Demo-regime ablation summary

All numbers in this section are ****demo-regime pipeline metrics**** measured on synthetic public-proxy tensors after short training schedules. They establish that reconstruction, segmentation, and motion terms are logged end-to-end; they are ****not**** estimates of clinical accuracy on licensed challenge cohorts.

Representative reconstruction PSNR on motion-relevant runs is approximately 19.1 dB with MAE approximately 0.69 under the compact fake-data regime. Motion-enabled runs log non-zero warp and image-cycle errors. Segmentation Dice on synthetic labels is low and class-imbalanced (right ventricle often near zero), consistent with under-trained demonstration schedules rather than a clinical ceiling.

****Table 1. Reconstruction and motion metrics (demo regime).****

Configuration	PSNR (dB)	MAE	Warp L1	Image-cycle	EF proxy
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Broadcast baseline	19.08	0.693	—	—	—
Per-frame, no motion weights	19.15	0.692	(9.7×10^{-4})	≈ 0	0.088
Per-frame + motion	19.15	0.695	(1.1×10^{-3})	(3.2×10^{-4})	0.081
Fusion (concatenation)	19.09	0.692	(5.0×10^{-4})	(1.1×10^{-4})	0.146
Fusion (HeartTTable-lite)	19.26	0.689	(1.3×10^{-3})	(7.6×10^{-4})	0.068

****Table 2. Segmentation Dice (demo regime).****

Configuration	Dice LV	Dice RV	Dice myocardium	Dice mean
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Broadcast baseline	0.227	≈0	0.278	0.168
Per-frame, no motion weights	0.014	≈0	0.495	0.170
Per-frame + motion	0.038	≈0	0.439	0.159
Fusion (HeartTTable-lite)	≈0	≈0	0.567	0.189
Phenotype task	0.053	≈0	≈0	0.018

****Phenotype diagnostics.**** Logged phenotype accuracy of 0.0 with area under the curve of 1.0 on a tiny synthetic split illustrates metric instability; we treat this as a pipeline check, not a scientific result. Columns corresponding to synthetic event AUC or concordance in demo logs are likewise diagnostics.

One-epoch demonstration checkpoints on recon- and phenotype-oriented configurations similarly show low Dice and mid-teens to ~19 dB PSNR.

5.2 Figures

- ****Fig. 1**** Pipeline schematic showing reconstruction, motion, segmentation, and lite-fusion roles.
- ****Fig. 2**** Reconstruction PSNR and MAE across ablations (****demo regime****).
- ****Fig. 3**** Warp, image-cycle, and EF proxies (****demo regime****).
- ****Fig. 4**** Dice heatmap across ablations (****demo regime****).
- ****Fig. 5**** Claim-boundary diagram versus the original clinical multimodal study.

Real ACDC cine volume curves, confidence intervals, and physical-space Hausdorff-95 remain to be completed after licensed mounts.

6. Discussion

The practical contribution of this work is a reproducible motion-consistent 4D training contract on public proxies, with documentation that prevents silent overclaim of the original AMI discrimination numbers. Image-cycle naming matters for peer review: reviewers familiar with registration literature will expect inverse consistency of flow fields; we instead regularize intensity cycles after warping.

HeartTTable-lite is intentionally under-powered relative to the original multimodal module. Building full three-class-token pairwise attention before real AMI data would optimize for a claim we cannot evaluate publicly.

Relative to whole-sequence memory segmentation^[5], unrolled joint recon–motion–seg^[3], and mesh/tetrahedra 4D recovery^{[7][8]}, our novelty claim is scoped to an auditable lightweight ReconSeg3D extension with explicit motion losses and fusion ablation boundaries—not leaderboard dominance.

****Limitations.****

1. Compact convolutional backbones are interface-compatible with, but not compute-matched to, paper-scale 3D vision transformers or full nnU-Net grids.
2. Sparse SA sampling is a slice-masking simulation, not vendor undersampled $\backslash(k\backslash)$ -space.
3. Global structural-similarity and subsampled Hausdorff-95 are approximate proxies on compact grids.

4. No nested cross-validation, calibration plots, or licensed challenge subject-level tables are reported yet.
5. Challenge-named local folders remain synthetic placeholders until licensed ACDC / MM-WHS / EMIDEC data are mounted.
6. Private AMI survival evaluation and the original five-year discrimination figures remain out of scope.
7. Clinical tabular vocabularies needed for a full HeartTTable-style study are not redistributed here.

7. Conclusions

We describe a methods-oriented, motion-consistent 4D extension of a ReconSeg3D-style pipeline for public cardiac MRI proxies, with HeartTTable-lite as a fusion ablation and an explicit refusal to claim private AMI clinical discrimination as ours. Measured demo-regime ablations demonstrate that reconstruction, segmentation, and motion metrics are logged end-to-end. Completing subject-level public-data tables after licensed dataset mounts—and, only with authorized private cohorts, survival evaluation—remain future work.

Data availability

Public benchmarks (ACDC, MM-WHS, EMIDEC) follow their respective licenses and access procedures (see repository data documentation). Official ACDC access requires institutional registration; this repository does not redistribute challenge volumes. Demo-regime metrics and figure sources are archived with the accompanying research report and figure bundle. No private AMI data are distributed with this repository. Challenge-named folders shipped for continuous integration contain synthetic placeholders only.

Code availability

<https://github.com/Coucou2016/ReconSeg3D-cardiac-rec>

Install dependencies listed in the repository requirements file (includes SciencePlots for figure regeneration). Demonstration figures can be regenerated with the repository figure script; the companion research report documents engineering inventory and process notes outside the scope of this manuscript.

References

- [¹]: Gao, Q. et al. 3D Spatiotemporal cardiac reconstruction for predicting MACE in acute myocardial infarction. **npj Digit. Med.** **9**, 325 (2026). <https://doi.org/10.1038/s41746-026-02449-0>
- [²]: Hammernik, K. et al. Deep learning for accelerated and robust MRI reconstruction. **MAGMA** (2024). <https://doi.org/10.1007/s10334-024-01173-8>
- [³]: Qian, P. et al. Unified Deep Learning for Simultaneous Cardiac Cine MRI Reconstruction, Motion Estimation and Segmentation. **IEEE ISBI** (2024). <https://doi.org/10.1109/ISBI56570.2024.10635390>
- [⁴]: Yuan, X. et al. 4D Myocardium Reconstruction with Decoupled Motion and Shape Model. **ICCV** (2023).
- [⁵]: Ye, M., Xin, B., Axel, L. & Metaxas, D. Continuous Spatio-Temporal Memory Networks for 4D Cardiac Cine MRI Segmentation. In **WACV** 9514–9524 (2025). Also [arXiv:2410.23191](https://arxiv.org/abs/2410.23191).

[^6]: Katzman, J. L. et al. DeepSurv: personalized treatment recommender system using a Cox proportional hazards deep neural network. *BMC Med. Res. Methodol.* **18**, 24 (2018). <https://doi.org/10.1186/s12874-018-0482-1>

[^7]: Chen, Y., Yang, J., Mercadier, D. S., Le, H. & Fua, P. MedTet: an online motion model for 4D heart reconstruction. *arXiv:2412.02589* (2024).

[^8]: Chen, Y. et al. End-to-end 4D heart mesh recovery across full-stack and sparse cardiac MRI. *Trans. Mach. Learn. Res.* (2026). Also *arXiv:2509.12090*.

Notes (nature-writing; not for submission body)

Axes: `task=manuscript` · `paper\_type=methods` · `language=en` · `journal=nature-family` (methods framing).

Claim–evidence map

Claim	Evidence	Status
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Motion-consistent 4D stack exists and logs metrics	Code + measured demo tables + Figs 2–3	Supported (demo regime)
Public clinical superiority vs baselines	Real ACDC folds	Needs evidence (incomplete)
HeartTTable parity / private AMI clinical AUC	—	Out of scope / rejected

Assumptions or missing inputs: licensed ACDC/MM-WHS/EMIDEC on disk; physical HD95; nested CV; private AMI for survival claims.

Why this structure: methods argument chain (problem → method → ablation → boundary); clinical multimodal paper deferred.

Figures

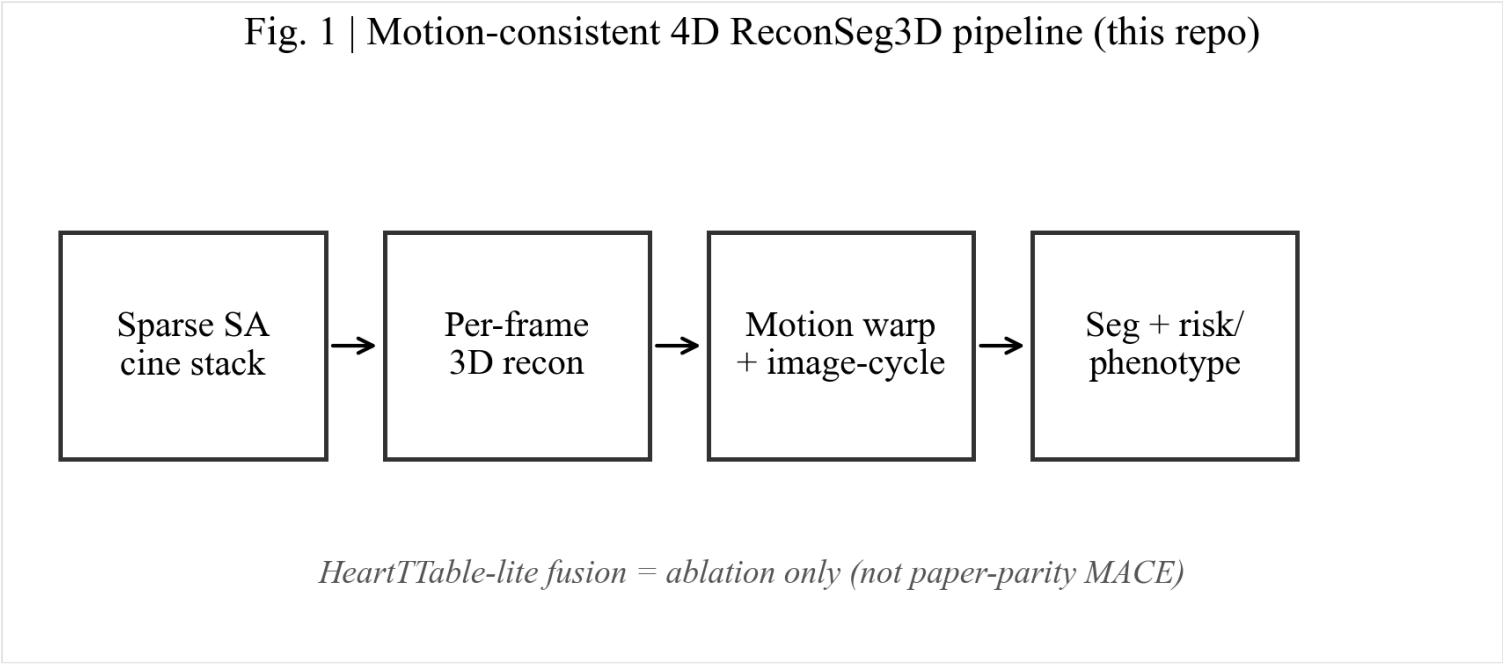


Figure 1
Pipeline schematic (SciencePlots).

Fig. 2 | Smoke ablation recon metrics (fake/demo data; not clinical)

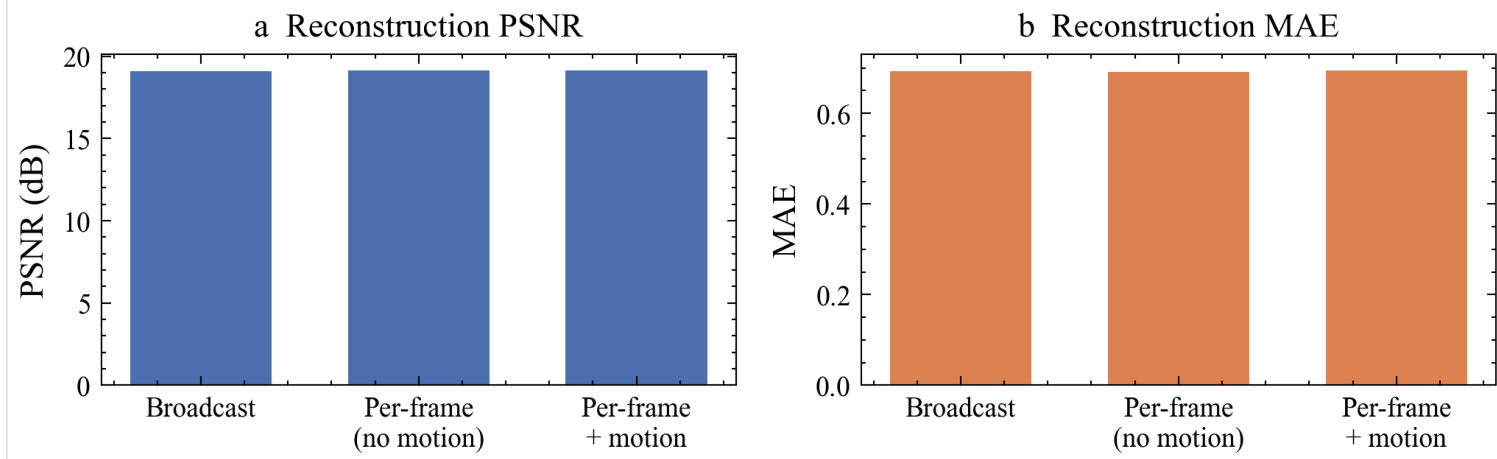


Figure 2
Reconstruction smoke ablation.

Fig. 3 | Motion / volume-curve proxies on smoke ablations (demo)

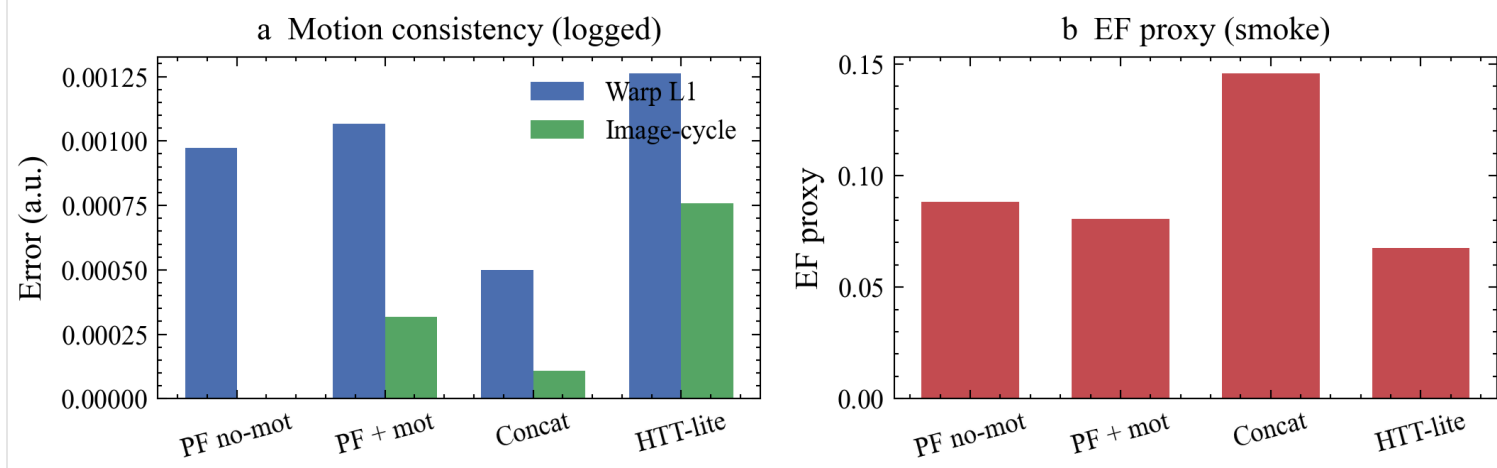


Figure 3
Motion proxies (demo).

Fig. 4 | Segmentation Dice heatmap (smoke / fake NIfTI; not clinical)

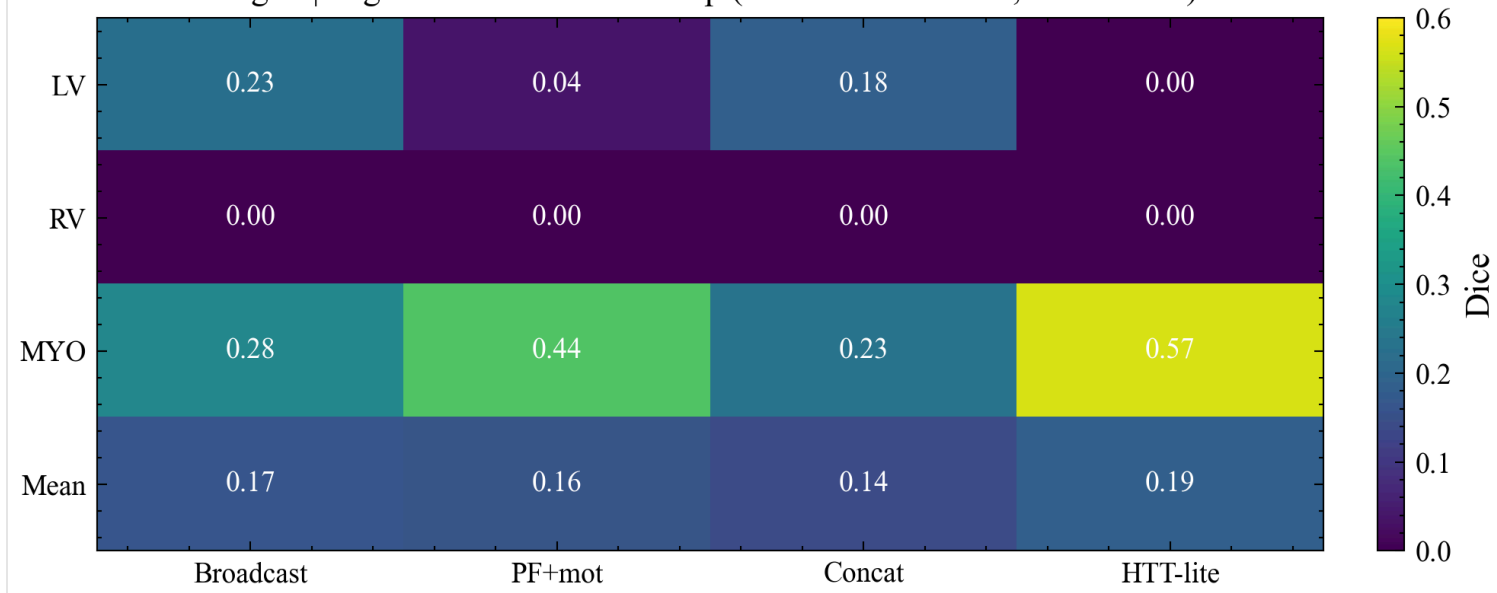


Figure 4

Dice heatmap (demo).

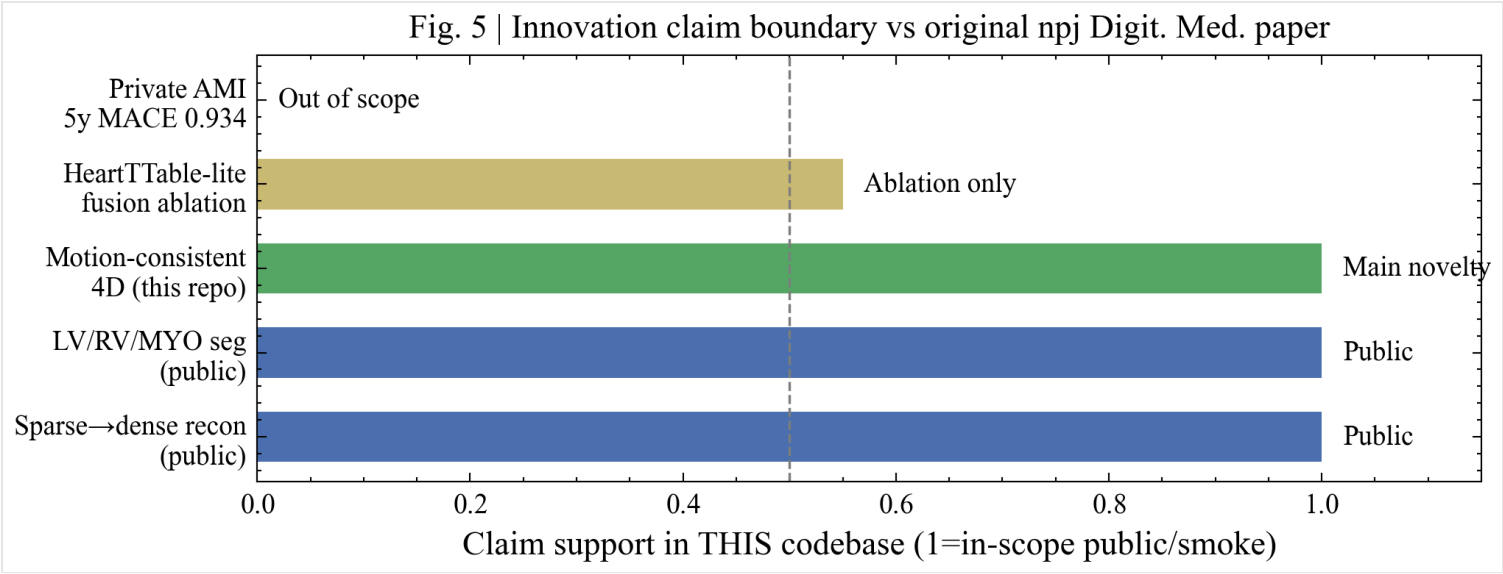


Figure 5

Claim boundary vs original paper.